

Practice Test II (Version 1)

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Trigonometry).

II) Show all work.

III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

IV) You will need a calculator for this test (be sure that it is in degree mode).

V) Each problem is worth four points for a total of 100 points.

Simplify the expressions below.

$$\textcircled{1} \left(\frac{\sin x}{\cos x} \right)^2 + \sec x \cos x$$

$$(\tan x)^2 + \frac{1}{\cos x} \cos x$$

$$\tan^2 x + 1$$

$$\boxed{\sec^2 x}$$

$$\textcircled{2} \frac{\csc x}{\sec x} + \cot x$$

$$\csc x \div \sec x + \cot x$$

$$\frac{1}{\sin x} \div \frac{1}{\cos x} + \cot x$$

$$\frac{1}{\sin x} \cdot \frac{\cos x}{1} + \cot x$$

$$\frac{\cos x}{\sin x} + \cot x$$

$$\cot x + \cot x$$

$$\boxed{2\cot x}$$

$$\textcircled{3} \frac{\cos^2 \theta + \cos \theta + \sin^2 \theta - 1}{1 + \cot^2 \theta - \sin \theta - \csc^2 \theta}$$

$$\frac{\sin^2 \theta + \cos^2 \theta + \cos \theta - 1}{\csc^2 \theta - \sin \theta - \csc^2 \theta}$$

$$\frac{1 + \cos \theta - 1}{-\sin \theta}$$

$$\frac{\cos \theta}{-\sin \theta}$$

$$\boxed{-\cot \theta}$$

Prove that the equations below are true by converting the left side to the right.

$$\textcircled{4} (\sin A + 1)(\sin A - 1) = -\cos^2 A$$

$$\sin^2 A - \sin A + \sin A - 1 =$$

$$\sin^2 A - 1 =$$

$$\sin^2 A - (\sin^2 A + \cos^2 A) =$$

$$\sin^2 A - \sin^2 A - \cos^2 A =$$

$$-\cos^2 A =$$

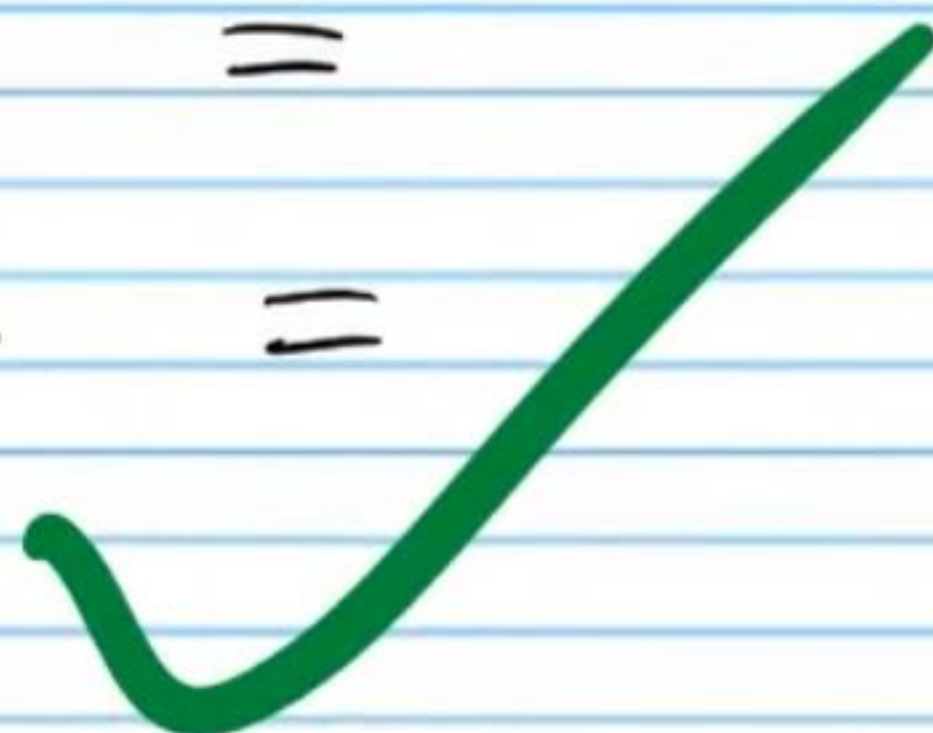


$$\textcircled{5} \sec B + \tan^2 B + \sec B = \sec^3 B$$

$$\sec B (\tan^2 B + 1) =$$

$$\sec B \sec^2 B =$$

$$\sec^3 B =$$



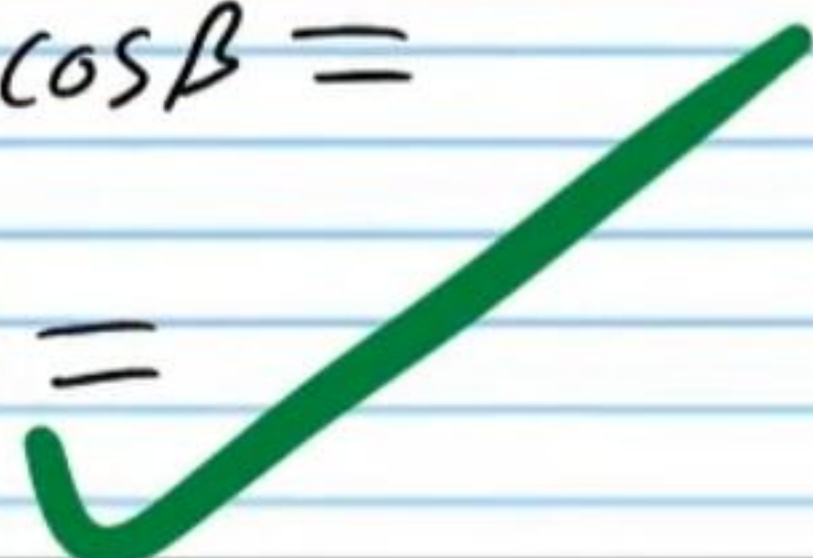
$$\textcircled{6} \frac{\sin \beta + \sec \beta}{\sec \beta \sin \beta} - \cos \beta = \csc \beta$$

$$\frac{\sin \beta}{\sec \beta \sin \beta} + \frac{\sec \beta}{\sec \beta \sin \beta} - \cos \beta =$$

$$\frac{1}{\sec \beta} + \frac{1}{\sin \beta} - \cos \beta =$$

$$\cos \beta + \csc \beta - \cos \beta =$$

$$\csc \beta =$$



$$\textcircled{7} \frac{\sin x}{1 + \sin x} + \frac{\sin x}{1 - \sin x} = 2 \tan x \sec x$$

$$\frac{\sin x (1 - \sin x)}{(1 + \sin x)(1 - \sin x)} + \frac{\sin x (1 + \sin x)}{(1 - \sin x)(1 + \sin x)} =$$

$$\frac{\sin x - \sin^2 x}{(1 + \sin x)(1 - \sin x)} + \frac{\sin x + \sin^2 x}{(1 - \sin x)(1 + \sin x)} =$$

$$\frac{\sin x - \sin^2 x + \sin x + \sin^2 x}{(1 + \sin x)(1 - \sin x)} =$$

$$\frac{2 \sin x}{1 - \sin x + \sin x - \sin^2 x} =$$

$$\frac{2 \sin x}{1 - \sin^2 x} =$$

$$\frac{2 \sin x}{\sin^2 x + \cos^2 x - \sin^2 x} =$$

$$\frac{2 \sin x}{\cos^2 x} =$$

$$\frac{2 \sin x}{\cos x \cos x} =$$

$$2 \left(\frac{\sin x}{\cos x} \right) \left(\frac{1}{\cos x} \right) =$$

$$2 \tan x \sec x =$$



⑧ Solve if $0 < \theta < 360^\circ$. Round to the nearest tenth.

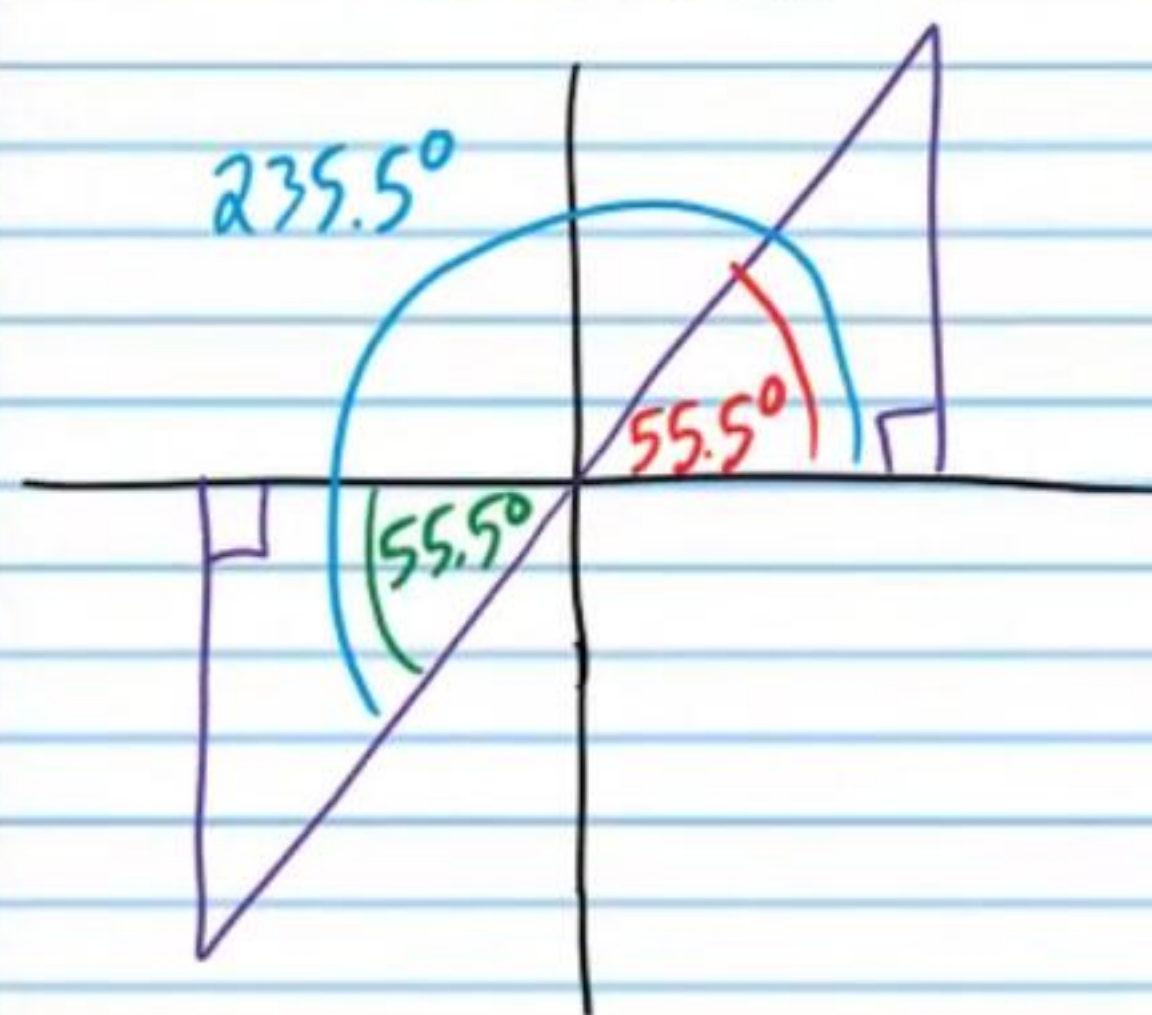
$$\tan \theta + 15 = 12 \tan \theta - 1$$

$$16 = 11 \tan \theta$$

$$\frac{16}{11} = \tan \theta$$

$$\theta = \tan^{-1}\left(\frac{16}{11}\right)$$

$$\theta \approx 55.5^\circ$$



$$\theta = 55.5^\circ, 235.5^\circ$$

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\cos^2 \alpha = 1 - \sin^2 \alpha$$

⑨ Solve if $0 \leq \alpha \leq 360^\circ$.

$$2 \sin \alpha = 8 \sin \alpha - 3 - \cos^2 \alpha - 2 \sin^2 \alpha$$

$$2 \sin \alpha = 8 \sin \alpha - 3 - (1 - \sin^2 \alpha) - 2 \sin^2 \alpha$$

$$2 \sin \alpha = 8 \sin \alpha - 3 - 1 + \sin^2 \alpha - 2 \sin^2 \alpha$$

$$2 \sin \alpha = 8 \sin \alpha - 4 - \sin^2 \alpha$$

$$\sin^2 \alpha - 6 \sin \alpha + 4 = 0$$

$$\text{Let } x = \sin \alpha$$

$$x^2 - 6x + 4 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(4)}}{2(1)}$$

$$x = \frac{6 \pm \sqrt{36 - 16}}{2}$$

$$x = \frac{6 \pm \sqrt{20}}{2}$$

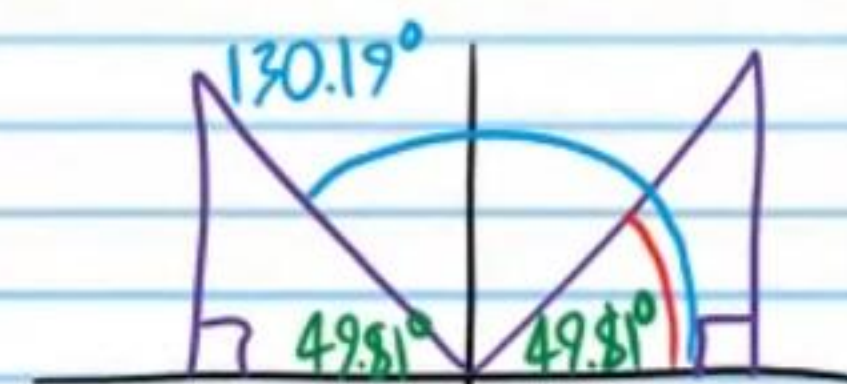
$$x = \frac{6 + \sqrt{20}}{2} \text{ or } x = \frac{6 - \sqrt{20}}{2}$$

$$x \approx 5.236 \text{ or } x \approx 0.7639$$

$$\sin \alpha \approx 5.236 \text{ or } \sin \alpha \approx 0.7639$$

$$\alpha \approx \sin^{-1}(0.7639)$$

$$\alpha \approx 49.81^\circ$$



$$\alpha \approx 49.81^\circ, 130.19^\circ$$

⑩ Find all values of W .

$$\sin W - \cos W = 0$$

$$\sin W - \cos W + \cos W = 0 + \cos W$$

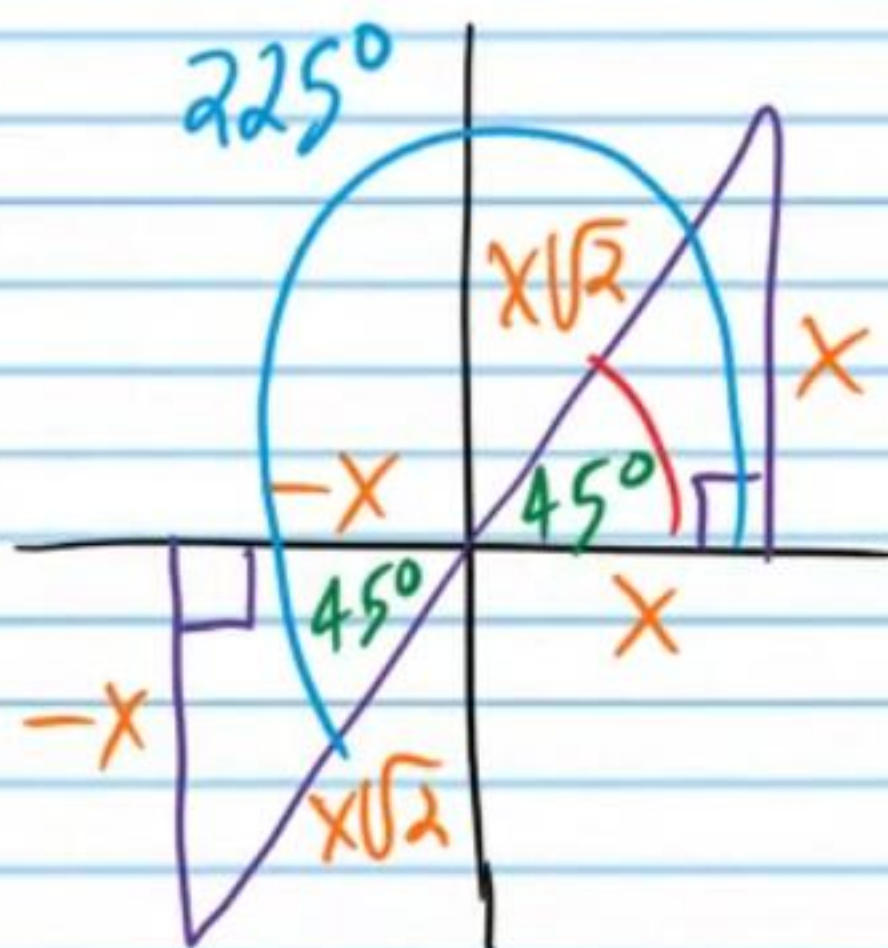
$$\sin W = \cos W$$

$$\frac{\sin W}{\cos W} = \frac{\cos W}{\cos W}$$

$$\tan W = 1$$

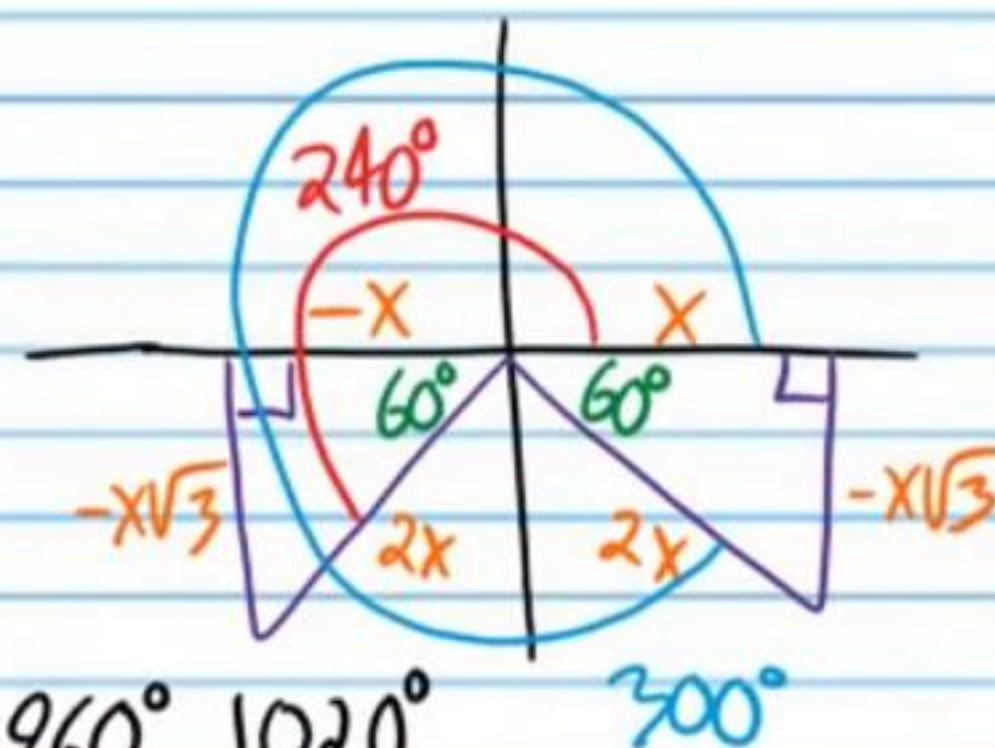
$W = 45^\circ + 360n$, n is any integer.

$W = 225^\circ + 360n$, n is any integer.



⑪ Solve if $0 < A \leq 360^\circ$.

$$\sin(3A) = -\frac{\sqrt{3}}{2}$$



$$3A = 240^\circ, 300^\circ, 60^\circ, 660^\circ, 960^\circ, 1020^\circ$$

$$\frac{3A}{3} = \frac{240^\circ}{3}, \frac{300^\circ}{3}, \frac{60^\circ}{3}, \frac{660^\circ}{3}, \frac{960^\circ}{3}, \frac{1020^\circ}{3}$$

$$A = 80^\circ, 100^\circ, 200^\circ, 220^\circ, 320^\circ, 340^\circ$$

⑫ Solve if $0^\circ \leq \alpha < 360^\circ$

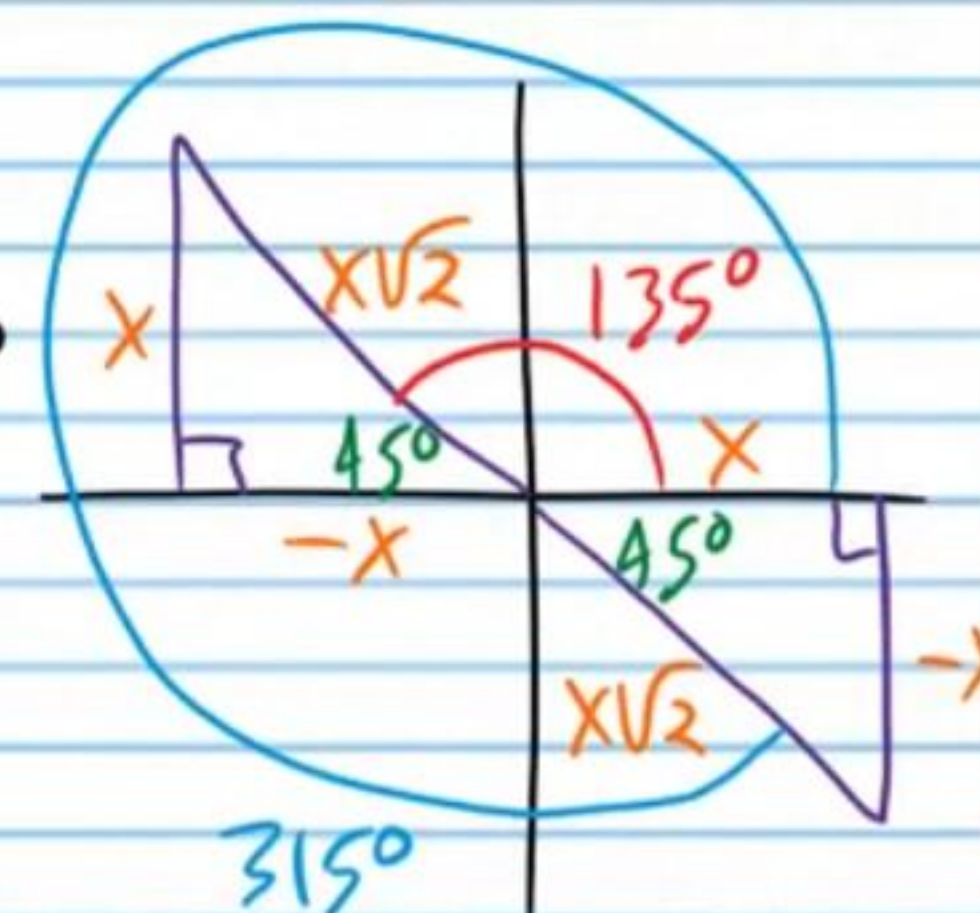
$$\cot\left(\frac{\alpha}{2} - 45^\circ\right) = -1$$

$$\frac{\alpha}{2} - 45^\circ = -45^\circ, 135^\circ, 315^\circ$$

$$\frac{\alpha}{2} = 0^\circ, 180^\circ, 360^\circ$$

$$\alpha = 0^\circ, 360^\circ, 720^\circ$$

$$\alpha = 0^\circ$$



⑬ Solve if $0 \leq \theta < 360^\circ$

$$\sin^2 \theta + 5 = \cos \theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$1 - \cos^2 \theta + 5 = \cos \theta$$

$$0 = \cos^2 \theta + \cos \theta - 6$$

$$\text{Let } x = \cos \theta$$

$$0 = x^2 + x - 6$$

$$0 = (x+3)(x-2)$$

$$x = -3 \quad \text{or} \quad x = 2$$

$$\cos \theta \neq -3$$

$$\cos \theta \neq 2$$

No solution

Find the values of the expressions below *without a calculator*.

$$\textcircled{14} \sin 255^\circ = \sin(210^\circ + 45^\circ) = \sin 210^\circ \cos 45^\circ + \sin 45^\circ \cos 210^\circ$$

$$= \left(-\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(-\frac{\sqrt{3}}{2}\right)$$

$$= -\frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{-1-\sqrt{3}}{2\sqrt{2}}$$

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$$

$$= \frac{(-1-\sqrt{3}) \cdot \sqrt{2}}{2\sqrt{2} \cdot \sqrt{2}}$$

$$= \frac{-\sqrt{2}-\sqrt{6}}{4}$$

$$= \boxed{-\frac{\sqrt{2}+\sqrt{6}}{4}}$$

$$\begin{aligned}
 (15) \sec 15^\circ &= \frac{1}{\cos 15^\circ} = \frac{1}{\cos(60^\circ - 45^\circ)} \\
 &= \frac{1}{\cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ} \\
 &= \frac{1}{\left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right)} \\
 &= \frac{1}{\frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}}} \\
 &= \frac{1}{\frac{1+\sqrt{3}}{2\sqrt{2}}} \\
 &= 1 \div \frac{1+\sqrt{3}}{2\sqrt{2}} \\
 &= 1 \cdot \frac{2\sqrt{2}}{1+\sqrt{3}} \\
 &= \boxed{\frac{2\sqrt{2}}{1+\sqrt{3}}}
 \end{aligned}$$

(16) Simplify the expression below. Do not use a calculator.

$$\cos 48^\circ \cos 40^\circ - \sin 48^\circ \sin 40^\circ = \cos(48^\circ + 40^\circ) = \boxed{\cos 88^\circ}$$

(17) Solve if $0 < \theta < 360^\circ$.

$$2\cos\theta = \sin(2\theta)$$

$$2\cos\theta = 2\sin\theta\cos\theta$$

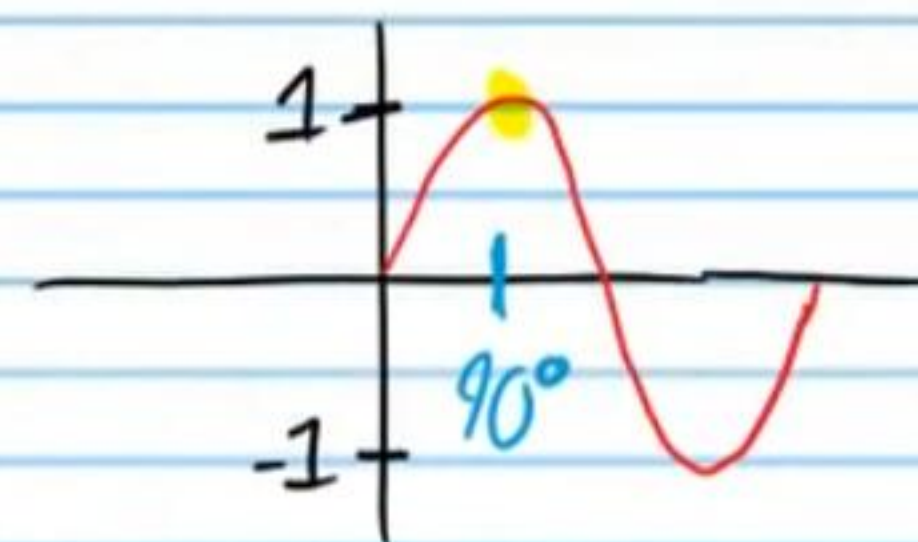
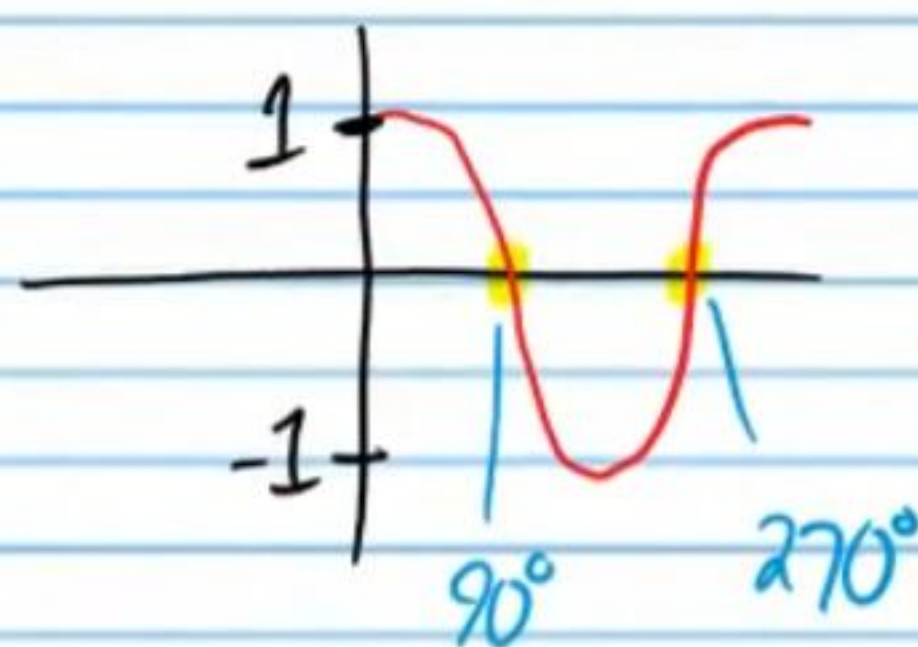
$$2\cos\theta - 2\cos\theta = 2\sin\theta\cos\theta - 2\cos\theta$$

$$0 = 2\sin\theta\cos\theta - 2\cos\theta$$

$$0 = 2\cos\theta(\sin\theta - 1)$$

$$2\cos\theta = 0 \quad \text{or} \quad \sin\theta - 1 = 0$$

$$\cos\theta = 0 \quad \text{or} \quad \sin\theta = 1$$



$$\theta = 90^\circ, 270^\circ \quad \text{or} \quad \theta = 90^\circ$$

$$\boxed{\theta = 90^\circ, 270^\circ}$$

⑮ Prove that the following equation is true by converting the left side to the right.

$$\frac{\cos(2B) - \sin^2 B - \cos^2 B}{-2} = \sin^2 B$$

$$\frac{\cos^2 B - \sin^2 B - \sin^2 B - \cos^2 B}{-2} =$$

$$\frac{-\sin^2 B - \sin^2 B}{-2} =$$

$$\frac{-2\sin^2 B}{-2} =$$

$$\sin^2 B =$$



⑯ Solve if $0 \leq x \leq 360^\circ$.

$$3\sin^2 x + \cos(2x) = 3 - \sin x$$

$$3\sin^2 x + 1 - 2\sin^2 x = 3 - \sin x$$

$$\sin^2 x + 1 = 3 - \sin x$$

$$\sin^2 x + \sin x - 2 = 0$$

$$\text{Let } y = \sin x$$

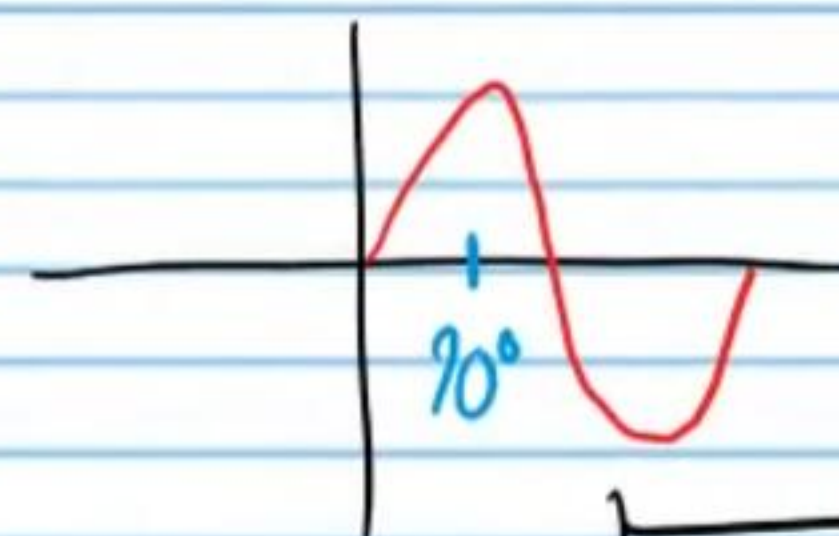
$$y^2 + y - 2 = 0$$

$$(y+2)(y-1) = 0$$



$$y = -2 \text{ or } y = 1$$

$$\sin x \neq -2 \text{ or } \sin x = 1$$



$$\boxed{x = 90^\circ}$$

Prove that the following equations are true by converting the left side to the right.

$$\textcircled{20} \quad 3\sin A + 8\sin(-A) - 2\sin A = -7\sin A$$

$$3\sin A + 8(-\sin A) - 2\sin A =$$

$$3\sin A - 8\sin A - 2\sin A =$$

$$-7\sin A =$$

$$\textcircled{21} \quad \frac{\cos(B+180^\circ)}{\cos B} = -1$$

$$\frac{-\cos B}{\cos B} =$$

$$-1 =$$

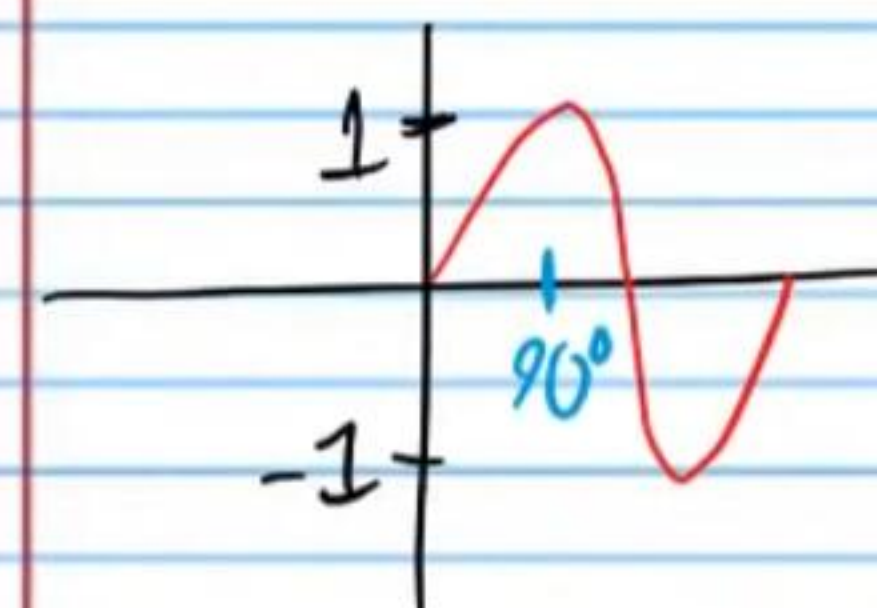
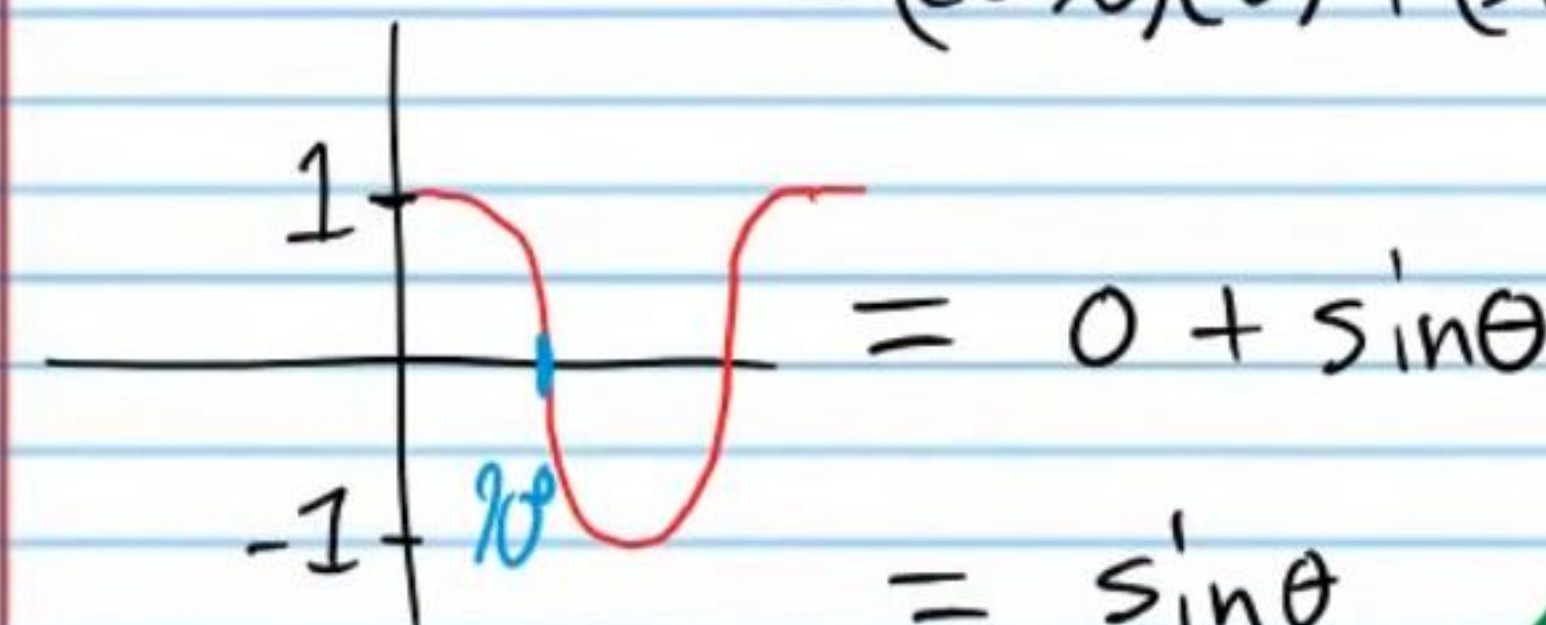
Use the sum formula to derive the following identity.

$$\textcircled{22} \quad \cos(\theta - 90^\circ) = \sin \theta$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\cos(\theta - 90^\circ) = \cos \theta \cos 90^\circ + \sin \theta \sin 90^\circ$$

$$= (\cos \theta)(0) + (\sin \theta)(1)$$

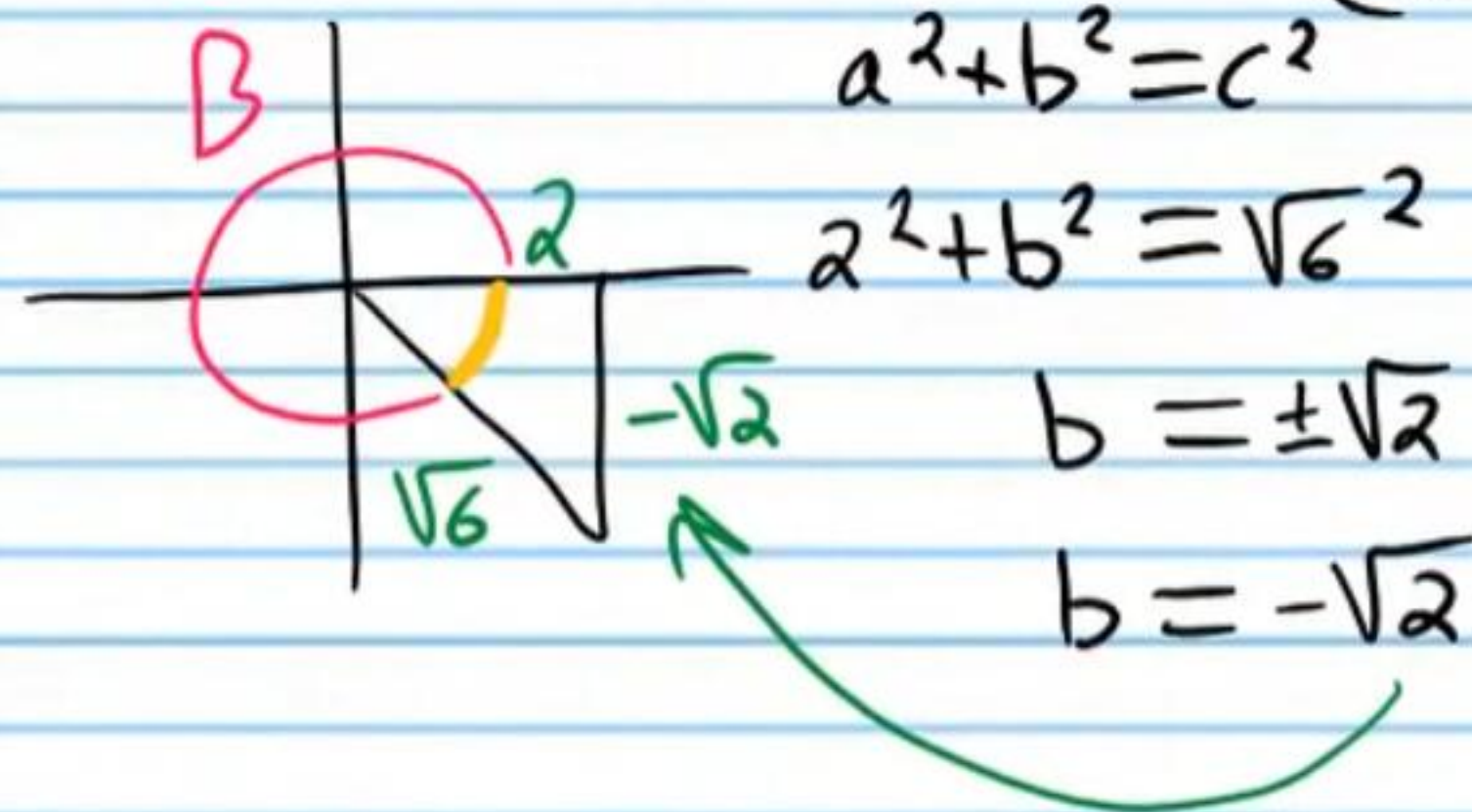


23) If B is in standard position, and it terminates in QIV ($270^\circ < B < 360^\circ$), find the value of each expression below if $\cos B = \frac{2}{\sqrt{6}}$.

$$\begin{aligned} \text{i) } \cos(2B) &= 2\cos^2 B - 1 = 2\left(\frac{2}{\sqrt{6}}\right)^2 - 1 = 2\left(\frac{4}{6}\right) - 1 \\ &= \frac{4}{3} - 1 \\ &= \boxed{\frac{1}{3}} \end{aligned}$$

ii) Find the value of $\sin(2B)$ using the double angle formula below.

$$\begin{aligned} \sin(2B) &= 2\sin B \cos B = 2\left(\frac{-\sqrt{2}}{\sqrt{6}}\right)\left(\frac{2}{\sqrt{6}}\right) = -\frac{4\sqrt{2}}{6} \\ &= \boxed{-\frac{2\sqrt{2}}{3}} \end{aligned}$$



24) If W is in standard position, and it terminates in QIII ($180^\circ < W < 270^\circ$), find the value of each expression below if $\cos W = -\frac{3}{5}$.

$$\text{i) } \cos\left(\frac{W}{2}\right) = \pm \sqrt{\frac{1+\cos W}{2}} = -\sqrt{\frac{1+(-\frac{3}{5})}{2}} = -\sqrt{\frac{\frac{2}{5}}{2}}$$

$$\begin{aligned} \frac{180^\circ}{2} &< \frac{W}{2} < \frac{270^\circ}{2} \\ 90^\circ &< \frac{W}{2} < 135^\circ \\ 180^\circ &< W < 270^\circ \end{aligned}$$

$$= -\sqrt{\frac{2}{10}}$$

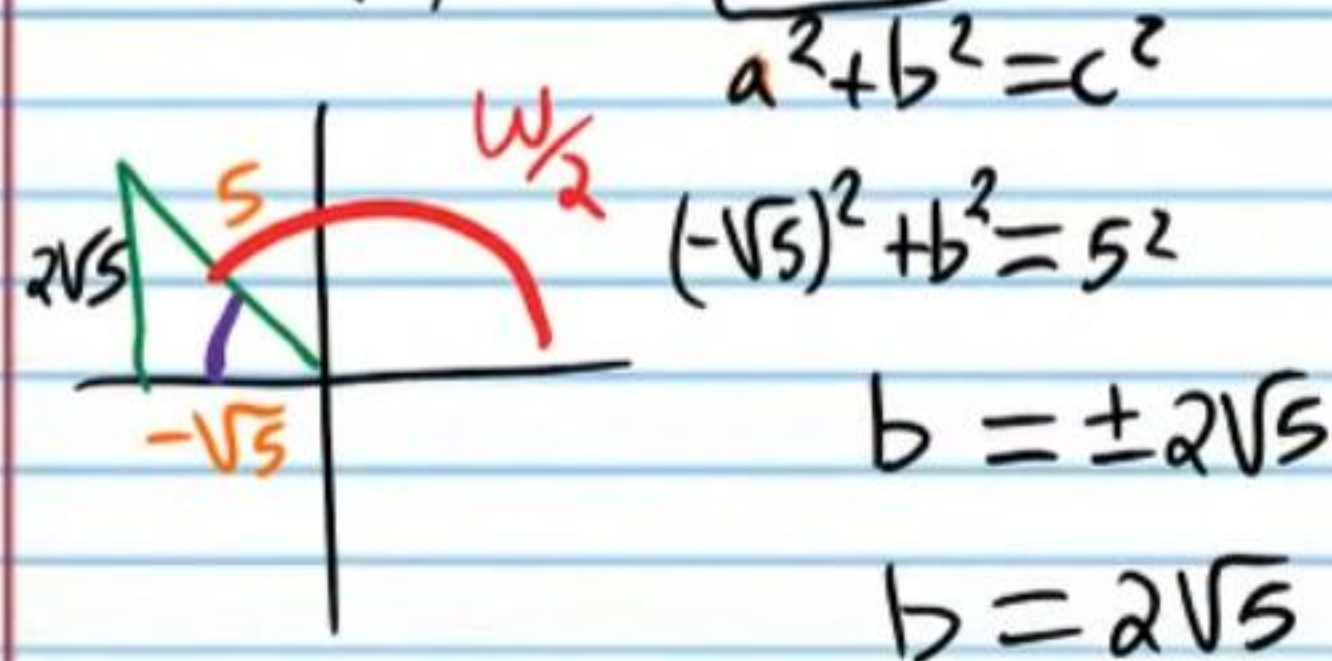
$$= -\sqrt{\frac{1}{5}}$$

$$= -\frac{\sqrt{1}}{\sqrt{5}}$$

$$= -\frac{1}{\sqrt{5}}$$

$$= \boxed{-\frac{\sqrt{5}}{5}}$$

$$\text{ii) } \sin\left(\frac{W}{2}\right) = \boxed{\frac{2\sqrt{5}}{5}}$$



$$a^2 + b^2 = c^2$$

$$(-\sqrt{5})^2 + b^2 = 5^2$$

$$b = \pm 2\sqrt{5}$$

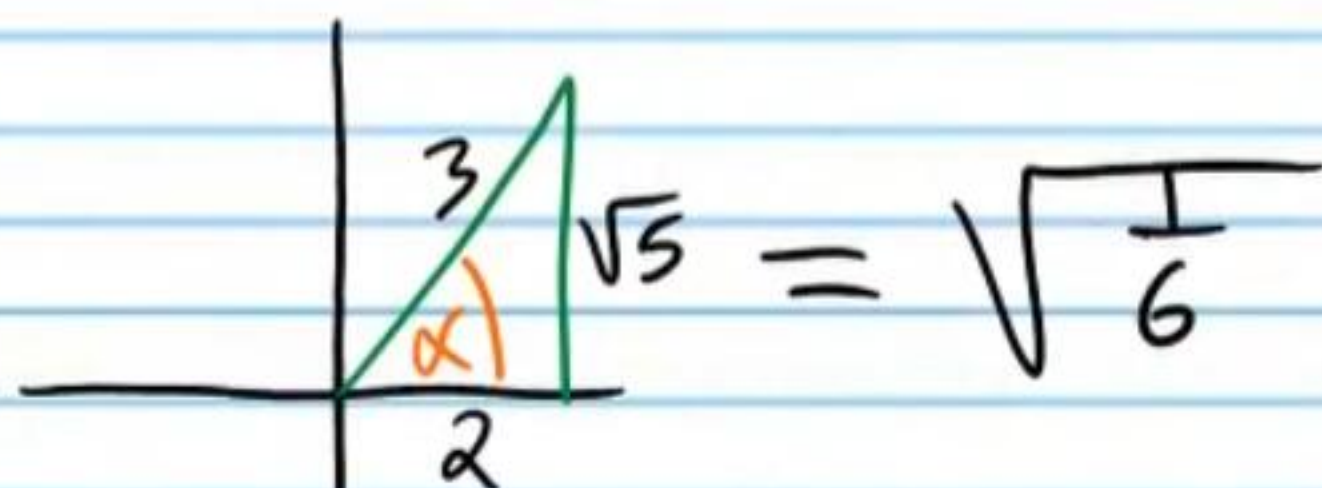
$$b = 2\sqrt{5}$$

25) If α is in standard position, and it terminates in QI ($0^\circ < \alpha < 90^\circ$), find the value of each expression below if $\sin \alpha = \frac{\sqrt{5}}{3}$.

$$i) \sin\left(\frac{\alpha}{2}\right) = \pm \sqrt{\frac{1 - \cos \alpha}{2}} = \sqrt{\frac{1 - \frac{2}{3}}{2}} = \sqrt{\frac{\frac{1}{3}}{2}}$$

$$0^\circ < \alpha < 90^\circ$$

$$0^\circ < \frac{\alpha}{2} < 45^\circ$$



$$a^2 + b^2 = c^2 = \frac{\sqrt{1}}{\sqrt{6}}$$

$$(\sqrt{5})^2 + b^2 = 3^2 = \frac{1}{\sqrt{6}}$$

$$b = \pm 2$$

$$b = 2 = \boxed{\frac{\sqrt{6}}{6}}$$

$$ii) \sec\left(\frac{\alpha}{2}\right) = \frac{6}{\sqrt{30}} = \frac{6\sqrt{30}}{30} = \boxed{\frac{\sqrt{30}}{5}}$$

